LOCAL CRITERIA FOR THE TOP FITTING IDEAL: GENERATION BY K ELEMENTS AT A PRIME

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Nakayama's lemma, which lets generation of the fibre by k elements spread to a localisation. Algebra, Lemma \ref{algebra-lemma-NAK}, cited at proof lines 1538-1540. Gives the equivalence of the last three conditions.
- Formation of Fitting ideals commutes with base change. Lemma \ref{lemma-fitting-ideal-basics}, cited at proof lines 1541-1543. Reduces the comparison with the non-containment condition to the case of a field.
- The Fitting ideal of a finite module over a field, where the module is free and the ideal is generated accordingly. Example \ref{example-fitting-free}, cited at proof lines 1545-1546. Settles the case of a field.

Source reference. Original lemma 07ZC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10142–10142. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07ZA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10140–10140. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Source reference. Original lemma 07ZB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10141–10141. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M be a finite R-module. Let $k \geq 0$. Let $\mathfrak{p} \subset R$ be a prime ideal. The following are equivalent*

(1) $Fit_k(M) \not\subset \mathfrak{p}$,
(2) $\dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p}) \leq k$,
(3) $M_\mathfrak{p}$ *can be generated by k elements over $R_\mathfrak{p}$, and*
(4) M_f *can be generated by k elements over R_f for some $f \in R$, $f \notin \mathfrak{p}$.*

Proof. By Nakayama's lemma (Algebra, Lemma 00DV) we see that M_f can be generated by k elements over R_f for some $f \in R$, $f \notin \mathfrak{p}$ if $M \otimes_R \kappa(\mathfrak{p})$ can be generated by k elements. Hence (2), (3), and (4) are equivalent. Using Lemma 07ZA part (3) this reduces the problem to the case where R is a field and $\mathfrak{p} = (0)$. In this case the result follows from Example 07ZB. □